

Project: House price Prediction

Dataset Used: Kc_house_data.csv

This project predicts house prices using multiple machine learning regression algorithms based on property features. The machine learning models can effectively predict house prices, and ensemble methods give better accuracy.

Project Goal: Train a module to predict the classes using Kc_house_data.csv dataset.

Technologies Used:

- **NumPy** : Numerical Operations
- **Pandas** : Data Handling
- **Scikit-Learn** : ML models & tools

Project Coverage:

Category	Skill Covered
Data preprocessing	Handling Dataset using Pandas
Create Learning Dataset	Select the required columns for passing as Input.
Training the module	Algorithms to train the module using the dataset.
Predict the result	Algorithms to predict the species class

Project covers data preprocessing, multiple ML models, prediction, and evaluation for iris classification. we are using different classes from the sklearn module to predict the species class with different algorithms as shown below.

1. Multiple Linear Regression

- Finds a relationship between multiple inputs and one output
- Fits a straight-line equation in multi-dimensions

```
from sklearn.linear_model import LinearRegression
regressor = LinearRegression()
print(regressor)

regressor.fit(X_train_scaled, y_train)
y_pred = regressor.predict(X_test_scaled)

print('\n'+ '-'*20+'R2 Score on the Test set'+ '-'*20)
print("{:.2f}".format(r2_score(y_test, y_pred)))
```

Multiple Linear Regression is a machine learning algorithm used to predict a continuous value using two or more input features.

2. Decision Tree Regression

- Divides data into regions
- Each region gives a constant predicted value
- Works like a flowchart

```
from sklearn.tree import DecisionTreeRegressor
regressor = DecisionTreeRegressor(random_state=0)
print(regressor)

regressor.fit(X_train_scaled, y_train)
y_pred = regressor.predict(X_test_scaled)

print('\n'+ '-'*20+ 'R2 Score on the Test set'+ '-'*20)
print("{:.2f}".format(r2_score(y_test, y_pred)))
```

Decision Tree Regression is a supervised learning algorithm used to predict continuous values by splitting data into smaller parts using decision rules (if-else conditions).

3. Random Forest Regression

- Creates random subsets of data (sampling)
- Builds a decision tree for each subset
- Each tree gives a prediction
- Final output = average of predictions

```
from sklearn.ensemble import RandomForestRegressor
regressor = RandomForestRegressor(n_estimators=100, random_state=0)
print(regressor)

regressor.fit(X_train_scaled, y_train)
y_pred = regressor.predict(X_test_scaled)

print('\n'+ '-'*20+ 'R2 Score on the Test set'+ '-'*20)
print("{:.2f}".format(r2_score(y_test, y_pred)))
```

Random Forest Regression is a supervised learning algorithm that predicts continuous values by using multiple decision trees and averaging their outputs

4. Support Vector Regression (SVR)

- Instead of minimizing total error, SVR tries to keep errors within a boundary (ϵ - epsilon)
- Focuses only on important data points called support vectors

```
from sklearn.svm import SVR
regressor = SVR(kernel='rbf')
print(regressor)

regressor.fit(X_train_scaled, y_train)
y_pred = regressor.predict(X_test_scaled)

print('\n'+ '-'*20+ 'R2 Score on the Test set'+ '-'*20)
print("{:.2f}".format(r2_score(y_test, y_pred)))
```

Support Vector Regression (SVR) is a supervised learning algorithm used to predict continuous values by fitting data within a specified error margin.

5. K-Nearest Neighbors (KNN)

- Finds K closest points to the new data
- Uses their values to make prediction

```
from sklearn.neighbors import KNeighborsRegressor
regressor = KNeighborsRegressor(n_neighbors=5)
print(regressor)

regressor.fit(X_train_scaled, y_train)
y_pred = regressor.predict(X_test_scaled)

print('\n'+ '-'*20+ 'R2 Score on the Test set'+ '-'*20)
print("{:.2f}".format(r2_score(y_test, y_pred)))
```

K-Nearest Neighbors (KNN) is a supervised learning algorithm that predicts output based on the nearest data points (neighbors).

6. Gradient Boosting Regression

- Combines many weak models (decision trees)
- Each tree improves the previous model
- Final prediction = sum of all models

```
from sklearn.ensemble import GradientBoostingRegressor
regressor = GradientBoostingRegressor(n_estimators=100, random_state=0)
print(regressor)

regressor.fit(X_train_scaled, y_train)
y_pred = regressor.predict(X_test_scaled)

print('\n'+ '-'*20+ 'R2 Score on the Test set'+ '-'*20)
print("{:.2f}".format(r2_score(y_test, y_pred)))
```

Gradient Boosting Regression is an ensemble learning algorithm that builds models step-by-step, where each new model corrects the errors made by the previous ones.

7. Ridge Regression

- Normal linear regression may overfit
- Ridge adds a penalty to keep coefficients small
- Helps improve model generalization

```
from sklearn.linear_model import Ridge
regressor = Ridge(alpha=1.0)
print(regressor)

regressor.fit(X_train_scaled, y_train)
y_pred = regressor.predict(X_test_scaled)

print('\n'+ '-'*20+ 'R2 Score on the Test set'+ '-'*20)
print("{:.2f}".format(r2_score(y_test, y_pred)))

from sklearn.metrics import mean_absolute_error
```

Ridge Regression is a type of linear regression that adds a penalty (L2 regularization) to reduce overfitting by shrinking large coefficients.

What you have learned:

In This project we have learned in the following.

- Learned to load and manage data using Pandas
- Selected relevant features and target (price)
- Performed train-test split
- Applied feature scaling using Scikit-learn
- Difference between regression and classification
- Concept of predicting continuous values

Conclusion:

This project demonstrates how machine learning can be used to predict house prices based on multiple property features. Data was loaded and processed using Pandas. Feature scaling improved model performance using Scikit-learn. Multiple regression algorithms were applied, including Linear Regression, Decision Tree, Random Forest, SVR, KNN, Gradient Boosting, and Ridge Regression. Models were evaluated using R^2 score and MAE.